

Direct Acoustic versus AUV-Assisted Hybrid Reporting for Subsea Pipeline Monitoring: A Matched End-to-End Evaluation

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Abstract

Subsea pipeline monitoring must deliver urgent alarms quickly while moving much larger inspection records over constrained underwater links. This creates a deployment choice: send all data through a prioritized acoustic network, or collect routine records with an autonomous underwater vehicle (AUV) already performing an inspection patrol. We compare these complete reporting architectures under identical traffic, geometry, deadlines, retry rules, and surface infrastructure. A scheduling-only control separates the benefit of alarm priority from the added effect of AUV collection. Finite campaigns and sustained operation are evaluated, with energy counted at equipment, patrol, and full-mission levels. In the finite campaign, AUV-assisted reporting reduced mean alarm delay by **76.1%** and acoustic demand, while lowering eventual delivery by **1.22 percentage points**. In sustained operation, direct priority maintained near-complete delivery, whereas about **15%** of status records missed their deadlines under the tested hybrid policy because records accumulated at gateways before pickup. Local energy savings became modest when communication was added to an existing patrol and disappeared when a full vehicle mission was attributed to collection. Direct priority is stronger for continuous completion and status timeliness; AUV-assisted reporting is valuable when alarm latency, gateway acoustic demand, and patrol reuse govern the deployment decision.

Keywords: subsea pipeline monitoring; AUV data collection; hybrid acoustic and optical communication; deadline-aware architecture; end-to-end delivery; underwater sensor networks; energy accounting

Abbreviations

The following abbreviations and traffic labels are used throughout the manuscript:

AoI	Age of Information
AUV	Autonomous Underwater Vehicle
BSN	Basic Sensor Node
CI	Confidence Interval
DDN	Data Dissemination Node (collection gateway)
FIFO	First-In, First-Out
HIGH	Alarm traffic with the strictest deadline
LOW	Routine bulk records eligible for delayed collection
MAC	Medium Access Control
MEDIUM	Status traffic with an intermediate deadline
PDR	Packet Delivery Ratio, as reported by prior studies
RDR	Record Delivery Ratio

Received:

Revised:

Accepted:

Published:

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Submitted to *J. Mar. Sci. Eng.* for

possible open access publication

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1. Introduction

Subsea pipeline monitoring produces two communication demands with different operational consequences. Pressure transients and acoustic-emission systems can generate compact alarms that must reach a surface team quickly enough to support intervention, whereas in-line and local inspection systems produce much larger records for condition assessment and maintenance planning [1–3]. Both streams depend on underwater links with limited capacity, long propagation times, and battery-powered equipment that is costly to access. A deployment must therefore protect urgent reporting without making routine inspection data unnecessarily expensive to move.

Two complete service chains address that problem. A direct service keeps every record on the acoustic path and uses priority scheduling to protect alarms. A hybrid service keeps the same acoustic access for urgent traffic but assigns routine records to short optical contacts with an autonomous underwater vehicle (AUV), which carries them until surface transfer. Acoustic links provide the required reach but at a comparatively low rate, whereas optical links provide high-rate service only at close range [4–6]. The operational decision is whether adding AUV pickup creates enough value in alarm response, acoustic demand, and submerged energy to justify the associated waiting, incomplete service, and vehicle cost.

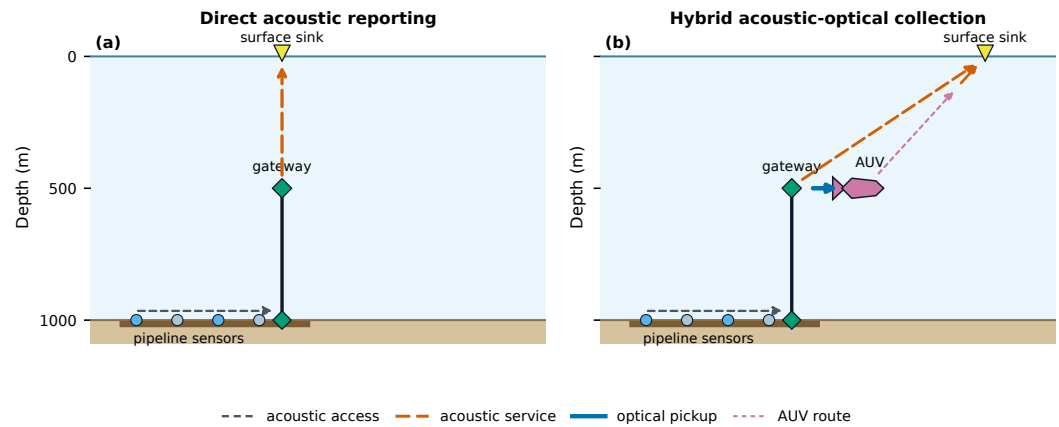


Figure 1. Operational alternatives evaluated in this study. Direct priority sends every gateway record acoustically to a surface station. Hybrid priority preserves acoustic access for alarms and aging status records while an AUV retrieves routine records optically and later transfers them to the surface. The alternatives share the same traffic, priorities, gateways, surface stations, and geometry.

Prior research on pipeline networks has established linear sensor deployments, AUV collection, priority-aware mobile sinks, and resurfacing or terminal-offload decisions [7–11]. Optimization and learning methods then select routes, visits, links, or schedules within an AUV-assisted system [12–16]. These studies establish the components required for mobile collection but leave the preceding deployment question open: how does a complete hybrid service compare with a correctly prioritized direct service when demand, geometry, deadlines, retry behavior, and surface infrastructure remain unchanged?

This study answers that question through a matched end-to-end evaluation from gateway storage to complete surface delivery. Direct FIFO first measures the effect of adding priority to the acoustic baseline. Direct priority and hybrid priority then share the same class ordering, so their difference isolates path allocation, vehicle pickup, carriage, and final transfer. Finite traffic that is allowed to drain measures eventual completion; continuous production over complete patrol cycles tests whether service remains sustainable. Energy is reported at three operational boundaries: inaccessible submerged equipment,

communication added to an existing patrol, and a mission commissioned for data collection.

This design yields three contributions:

- a matched comparison that isolates scheduler value and the additional effect of AUV-assisted service;
- an end-to-end assessment that distinguishes eventual completion from continuous deadline performance and locates retained data along the service chain;
- a deployment-aware energy assessment that separates local equipment savings, patrol reuse, and the full cost of a dedicated vehicle mission.

The evaluation follows three questions. **RQ1** asks what priority scheduling changes and what additional trade-off hybrid pickup creates. **RQ2** asks how link quality, fallback timing, and continuous production shift that trade-off and where the active service constraint appears. **RQ3** asks under which energy boundary hybrid collection produces a deployment benefit. Section 2 positions these questions against the literature. Sections 3 and 4 define the alternatives and evaluation method, Section 5 presents the evidence, and Section 6 translates it into a deployment decision.

2. Related Work and Positioning

2.1. Subsea Pipeline Monitoring and Mobile Collection

Pipeline-monitoring research connects integrity sensing, linear-network communication, and mobile inspection. Pressure-transient, acoustic-emission, and in-line inspection studies establish the complementary information needed for leak detection and structural assessment [1–3]. At the network level, Jawhar et al. [7] described an AUV-assisted pipeline architecture, Ahmed et al. [9] evaluated collection in a linear sensor network, and Goyal et al. [10] used mobile sinks for priority-based leak-monitoring traffic. Zheng et al. [8] extended the service endpoint to the surface by jointly considering the AUV tour and resurfacing decision. Across these works, topology, collection points, and the endpoint that defines delivery are central architectural choices.

Mobile service introduces waiting and vehicle motion in exchange for reduced underwater transmission. Janardanan Kartha and Jacob [17] evaluated delay, delivery, energy, and network lifetime for mobility-assisted acoustic collection, while Hollinger et al. [18] connected robotic data collection with measured channel structure and medium access. Queueing and information-value studies then separated eventual delivery from freshness and revisit time [12,19–22]. Operational studies broaden the boundary further: resident-AUV routing includes docking and battery constraints [23], and terminal-offload planning retains a final service phase until data recovery or budget exhaustion [11]. This broader view motivates evaluating the complete record journey from gateway to surface and distinguishing patrol reuse from a mission created specifically for data collection.

2.2. Priority-Aware and Hybrid Underwater Service

Acoustic and optical links occupy complementary operating regimes. Acoustics provide long reach at a comparatively low rate; optics provide high-rate local service but require close contact and suitable alignment. Event simulation, hybrid routing, and multimodal collection studies have combined these regimes for reliability, delay, and energy management [6,24–28]. Weng et al. [29] demonstrated three-dimensional optical tracking and seafloor-platform data retrieval in sea trials, establishing the physical plausibility of local AUV pickup. A complete hybrid service must additionally account for queue residence, vehicle carriage, and the final transfer to a surface endpoint.

Recent optimization and learning methods determine how such an AUV-assisted service should operate. Liu et al. [30] combine urgent forwarding with value-aware retrieval;

Bu et al. [13] jointly select an acoustic or optical mode and an AUV trajectory; and Song et al. [14], Cao et al. [15], and Wei et al. [16] optimize motion, channel-aware collection, information freshness, or energy. Lee et al. [31] extend scheduling to the buoy endpoint through inter-AUV relay service. These methods optimize control decisions within an AUV-assisted architecture.

2.3. From Within-Architecture Optimization to Architecture Selection

The sensing, routing, optical-contact, scheduling, and trajectory components required for mobile collection are well established. The remaining deployment question is comparative: what operational value does the complete hybrid chain create relative to a priority-based direct chain under the same demand, geometry, deadlines, retry behavior, and surface infrastructure? Table 1 positions representative studies by the decision they optimize, the endpoint they measure, and the energy boundary they adopt.

Table 1. Representative underwater collection and pipeline-monitoring studies, ordered by publication year and positioned by decision, evaluation endpoint, and energy boundary.

Study	Year	Architecture / decision	Evaluation criterion	Energy boundary
Janardanan Kartha and Jacob [17]	2015	Direct/multihop versus mobile-sink acoustic collection	PDR; queue; delay to mobile sink	Sensor energy /lifetime; mobile-sink endpoint
Zheng et al. [8]	2017	Pipeline AUV tour and resurfacing	Mean reporting delay to surface	Route and resurfacing focus
Wang et al. [24]	2017	Hybrid optical-acoustic versus acoustic network	Connectivity; loss; end-to-end delay	Node energy balance and lifetime
Fang et al. [12]	2021	Multi-AUV limited-service collection	Peak AoI; stationary queue	Patrol and vertical-motion energy
Liu et al. [30]	2022	Within-architecture urgent forwarding and AUV retrieval	Value, timeliness, and energy efficiency	Node communication and AUV route
Ahmed et al. [9]	2022	AUV routing on a linear pipeline	PDR; end-to-end delay; throughput	Traffic-load evaluation
Bu et al. [13]	2023	Within-architecture trajectory and acoustic/optical selection	Importance- and size-weighted AoI	Node transmission and AUV-motion energy
Zhu et al. [28]	2024	Priority- and void-aware hybrid routing	Priority; PDR; delay	Forwarding energy; network lifetime
Kasparavičiūtė et al. [23]	2025	Resident-AUV route and reprioritization	Route time; priority completion	Battery threshold; docking and recharge
Wei et al. [16]	2025	Channel-aware AUV path planning	PDR; collection rate under loss	Node communication and AUV-motion energy
Weng et al. [29]	2025	Optical AUV tracking and seafloor-platform pickup	Link alignment; sea-trial retrieval	Tracking-energy objective
Lee et al. [31]	2026	Within-architecture collection/relay scheduling to a buoy	Buffer-coupled peak AoI; delivered packets	End-to-end relay and AoI focus
Zhang et al. [11]	2026	AUV-USV inspection with terminal offload	Final buffer; upload success; mission time	Motion-energy proxy; terminal phase included
This study	–	Direct-versus-hybrid architecture selection under a prescribed patrol	Deadlines; completion; backlog	Submerged; incremental-patrol; dedicated-mission

This study operates at the architecture-selection level. Direct FIFO and direct priority first isolate scheduler value. Direct priority and hybrid priority then use the same order-

ing, so their comparison isolates routine-data path allocation and AUV service. Finite and continuous experiments distinguish eventual completion from sustained operation, and three energy accounts distinguish submerged infrastructure, an existing patrol, and a dedicated collection mission. Prior work establishes the feasibility of each component; the following sections compare the two complete service chains under one common operational workload.

3. Architecture and Decision Problem

3.1. Operational Boundary and Record Lifecycle

Figure 1 defines the two gateway-to-surface service chains. Seabed sensors and relays first bring alarm and status records to eight elevated collection gateways. High-volume inspection instruments are represented at the gateways because the comparison begins where routine records enter the surface-delivery service. A record is delivered only after complete reassembly at a surface station. Gateway arrival and AUV pickup remain intermediate states.

Direct reporting sends every gateway record acoustically to a reachable surface station. Hybrid reporting preserves the acoustic path for urgent records but assigns routine records to scheduled optical pickup. A successful contact moves the record into the AUV carry queue, and the record remains incomplete until the vehicle finishes the modeled surface transfer. Both alternatives use the same pipeline, gateways, surface stations, offered traffic, class priorities, and retry behavior. The comparison therefore changes the service path, not the workload or supporting infrastructure.

3.2. Compared Service Configurations

Three configurations separate two effects that are often confounded in an architecture comparison. Table 2 identifies the role of each configuration. The transition from direct FIFO to direct priority measures the value of scheduling urgent records first. The transition from direct priority to hybrid priority is the primary architecture comparison: it measures the additional effect of optical pickup, vehicle carriage, and final surface service while holding class ordering constant.

Table 2. Service configurations and their role in the controlled comparison.

Configuration	Gateway scheduling	Routine-record path	Analytical role
Direct FIFO	Arrival order	Acoustic surface service	Scheduling control
Direct priority	HIGH, MEDIUM, LOW	Acoustic surface service	Primary direct baseline
Hybrid priority	HIGH, MEDIUM, LOW	Optical pickup and AUV carriage	Evaluated architecture

The workload distinguishes alarms (HIGH), status updates (MEDIUM), and routine inspection records (LOW). Alarms have the strictest deadline and retain prompt acoustic access in both priority configurations. Status records receive an initial opportunity for AUV pickup in hybrid service and later fall back to the acoustic path. Routine records remain acoustic in direct service and use AUV pickup in hybrid service. This class mapping makes the intended operational exchange explicit: scheduled collection can protect alarm service and reduce gateway transmission, but it introduces waiting and an additional completion stage for less urgent data.

3.3. Decision Criteria

The architecture is judged on complete surface delivery, alarm and status timeliness, acoustic transmission demand, the location of retained records, and energy under a stated deployment boundary. Two observation modes are required. Finite generation followed by queue drain measures whether records eventually complete after demand stops. Continuous production over repeated patrol cycles measures whether the service repeatedly meets deadlines while new records keep arriving. The energy analysis then distinguishes equipment that is difficult to access, communication added to a patrol that already exists, and a mission commissioned specifically for collection. With the alternatives and criteria defined, the following sections test them under a matched operational workload.

4. Materials and Methods

4.1. Reference Deployment and Workload

The reference corridor contains 180 nominal positions separated by 70 m over 12.6 km, consistent with role-based linear pipeline architectures [7,9]. Every fifth position acts as a relay, giving 144 traffic-generating sensors and 36 relays. Eight relay positions connect to elevated gateways at 500-m depth, and 15 surface stations provide direct or terminal reception opportunities along the corridor. This dense fixed layout gives direct reporting frequent surface receivers and forms a demanding baseline for the added value of vehicle collection.

The AUV follows a prescribed smooth out-and-back patrol through the gateways and intervening surface stations. It travels at 6.09 km/h, within the operating range of survey-class AUVs [32], and completes one patrol cycle in approximately 17,560 s. Records arrive at a mean rate of 0.05 record/s with probabilities 0.10, 0.30, and 0.60 for HIGH, MEDIUM, and LOW traffic. Their payloads and deadlines are 64 bytes and 120 s, 256 bytes and 600 s, and 10 or 25 kB and 30,000 s, respectively. Matched random streams preserve arrival times, classes, and sources across compared configurations.

4.2. Service and Communication Representation

Each gateway stores up to 100 records. In hybrid service, HIGH records gain acoustic access at age 2 s, while MEDIUM records request fallback at age 120 s if they still await pickup. A shared one-credit token bucket regulates that fallback. Its nominal refill of 0.015 record/s equals the mean offered rate of the complete MEDIUM stream, leaving no nominal reserve before scheduled pickup is considered. Sensitivity tests at 60, 120, and 240 s examine how long status records should wait before requesting the acoustic path.

The event-driven model represents propagation delay, spatial interference, distance-dependent reception, fragmentation, retries, reassembly, optical contact, vehicle storage, and final surface service. Acoustic service uses a 31.2-kbit/s rate and a synthetic reference margin of 52 dB, with 58- and 64-dB controls. Optical pickup uses a 30-m contact region and a 2.5-Mbit/s clear-water reference. Published acoustic-modem, optical-modem, attenuation, and vehicle-power sources provide engineering anchors [4,5,33–37]. The exact link, framing, retry, pointing, and power functions are documented in Appendix A; the main analysis uses the same functions and service limits for every alternative.

4.3. Deadline and Energy Measures

Deadline performance includes only records old enough to receive their complete deadline window. For class C with deadline d_C and run end T , let N_C^{obs} count records

generated by $T - d_C$, and let D_C^{on} count members delivered to a surface station by their individual deadline. The miss ratio is

$$\hat{m}_C = 1 - \frac{D_C^{\text{on}}}{N_C^{\text{obs}}}, \quad N_C^{\text{obs}} > 0. \quad (1)$$

This denominator prevents recently generated records from being labeled late before their deadline has elapsed. Mean delay is calculated for delivered records and is always interpreted with the corresponding delivery count. The reference service targets are 1% misses for HIGH traffic and 5% for MEDIUM traffic.

Energy is reported through three deployment accounts. Let E_{sub} contain active and idle energy of submerged sensors, relays, and gateways; E_{sink} surface-station communication energy; $E_{\text{comm,AUV}}$ AUV communication energy; and $E_{\text{AUV}}^{\text{op}}$ propulsion and hotel load. The incremental account represents communication added to an unchanged inspection patrol, and the dedicated account assigns the full vehicle operation to collection:

$$\begin{aligned} E_{\text{inc}} &= E_{\text{sub}} + E_{\text{sink}} + E_{\text{comm,AUV}}, \\ E_{\text{ded}} &= E_{\text{inc}} + E_{\text{AUV}}^{\text{op}}. \end{aligned} \quad (2)$$

Direct reporting has zero AUV terms. Submerged, incremental-patrol, and dedicated-mission energy are reported separately so that a local communication saving cannot be mistaken for a complete mission-energy saving.

4.4. Matched Experiments and Observation Horizons

Table 3 separates the primary architecture comparison from three controls. The reference contains 20 matched runs for direct FIFO, direct priority, and hybrid priority. The acoustic-margin study compares direct and hybrid priority at 52, 58, and 64 dB using 10 common seeds per margin. The status-fallback study evaluates hybrid age gates of 60, 120, and 240 s with 20 runs per setting. Continuous service is evaluated for both architectures and two routine-record sizes over five post-warm-up patrol cycles. Within every comparison, matched random streams preserve arrival times, class labels, and sources; each service then develops its own channel and queue history.

Table 3. Executed evaluation matrix. Finite cases generate traffic for 4000 s and then drain until $T = 23,450$ s; continuous cases keep production active through five complete post-warm-up patrol cycles. Counts are per configuration unless a matched set is stated.

Experiment	Arrival mode	Controlled factor	Runs
Reference comparison	Finite + drain	25-kB LOW; direct FIFO, direct priority, hybrid priority	20 triplets
Acoustic margin	Finite + drain	$M = 52, 58, 64$ dB; direct priority versus hybrid priority	10 pairs/ M
MEDIUM age gate	Finite + drain	Hybrid $\tau_M = 60, 120, 240$ s	20 each
Continuous service	Continuous	10- and 25-kB LOW; five post-warm-up patrol cycles	5 each

The finite mode stops generation after 4000 s and drains beyond one complete patrol cycle. It therefore measures eventual surface completion after demand has ceased. The continuous mode remains active through $T = 91,798.03$ s and samples storage at the exact end of each patrol cycle. An ordinary least-squares trend is fitted to the five cycle-end storage values for each run. Exact cycle boundaries avoid interpreting regular within-

patrol queue oscillation as persistent accumulation and describe the observed five-cycle behavior [38].

4.5. Outcomes and Statistical Comparisons

Record delivery ratio (RDR) is the number of records completely reassembled at a surface station divided by the number generated. Deadline ratios use the mature groups defined in Equation (1). Packet-lifecycle traces connect generation with the first complete surface acknowledgement, and the terminal ledger separates seabed-access loss, deadline expiration, admission loss, ongoing service, gateway storage, and AUV storage. These states distinguish a path that eventually drains from a service that keeps pace with continuous demand.

Acoustic transmitted bytes include headers, hops, and retransmissions. Normalized acoustic attempt load sums transmission time across all acoustic senders and divides by the observation horizon; concurrent spatial links contribute independently. The energy outcomes use the three boundaries in Equation (2). Together, these measures connect application outcomes to the service mechanism that produces them.

For the reference comparison, the primary effect is the paired within-seed hybrid-minus-direct-priority difference; direct priority minus direct FIFO measures scheduler sensitivity. Means and 95% confidence intervals use paired differences or independent run-level values with the Student distribution [38]. A paired percentile bootstrap checks the direction of bounded RDR and deadline effects. Margin tests use paired intervals at each setting, fallback tests use run-level intervals with pooled counts as an event-level audit, and the five-run continuous study serves as a diagnostic assessment of the four tested operating points.

4.6. Event Accounting and Reproducibility

Each experiment follows every record through gateway arrival, acoustic service, optical pickup, AUV carriage, and complete surface delivery while representing vehicle motion, spatial interference, deadlines, retries, and energy. The implementation uses ns-3.40 with AquaSim-NG [39,40]. Conservation checks assign every generated record to delivery, expiration, access loss, ongoing service, or storage. The supplementary package contains the simulator, pinned execution environment, experiment commands, run-level outputs, logs, and analysis code. The development repository is available at <https://github.com/Aissata98/ns3-new>.

5. Results

5.1. Hybrid Collection Accelerates Alarms but Weakens Status Service

The finite comparison reveals two distinct effects. Priority scheduling alone reduced mean alarm delay from 20.7 to 14.4 s without changing final delivery: direct FIFO and direct priority both delivered approximately 99.80% of records. Assigning routine records to the AUV produced the larger latency effect. Hybrid priority reduced mean alarm delay from 14.4 to 3.45 s, a 76.1% reduction, while acoustic transmitted bytes fell by 11.3%. These gains arise from two distinct mechanisms: serving urgent traffic first and moving routine traffic off the gateway acoustic path.

Hybrid collection did not improve final delivery. Its finite-case RDR was 98.59%, compared with 99.80% for direct priority, a matched reduction of 1.22 percentage points with a 95% interval from 0.47 to 1.96 points. Status misses increased from 0.58% to 4.67%, and mean delivered-status delay increased from 13.9 to 233.8 s. Table 4 therefore describes a conditional exchange: a small eventual-completion penalty and slower status service pur-

chase much faster alarms and lower acoustic demand. Every generated record remained accounted for as delivered, expired, lost during access, in service, or stored.

Table 4. Finite-traffic reference results for 25-kB routine records at $\lambda = 0.05$ records/s, followed by queue drain and averaged over 20 common seeds. The primary effect is hybrid priority minus direct priority; percentage effects are percentage points.

Metric	Direct FIFO	Direct priority	Hybrid priority	Primary effect [95% CI]
<i>Completion and deadline performance</i>				
End-to-end RDR (%)	99.80	99.80	98.59	−1.22 [−1.96, −0.47]
HIGH deadline misses (%)	1.18	0.24	0.22	−0.02 [−0.71, 0.67]
MEDIUM deadline misses (%)	0.43	0.58	4.67	+4.09 [1.67, 6.50]
<i>Delivered-record delay</i>				
HIGH conditional delay (s)	20.7	14.4	3.45	−10.99 [−13.93, −8.06]
MEDIUM conditional delay (s)	19.1	13.9	233.8	+219.9 [201.1, 238.8]
LOW conditional delay (s)	33.7	34.7	5924	+5890 [5640, 6139]
<i>Acoustic demand</i>				
Acoustic transmitted bytes (MB)	4.83	4.82	4.28	−0.54 [−0.64, −0.44]
Normalized acoustic attempt load (%)	5.28	5.27	4.68	−0.59 [−0.71, −0.48]
<i>Deployment energy accounts</i>				
Submerged-device energy (kJ)	11.99	11.99	2.40	−9.59 [−10.13, −9.05]
Incremental-patrol energy (kJ)	12.90	12.89	11.95	−0.94 [−1.16, −0.72]
Dedicated-mission energy (MJ)	0.0129	0.0129	5.324	+5.311 (descriptive)

5.2. Link Quality and Fallback Timing Define the Trade-off

The alarm advantage depends strongly on the direct acoustic link. Increasing the synthetic margin from 52 to 58 dB improved direct service and reduced the hybrid alarm-delay advantage from 11.52 to 1.51 s; the same 1.51-s difference remained at 64 dB. The acoustic-byte reduction contracted from 0.60 to 0.011 MB. Hybrid RDR remained limited by status expiration, while the matched RDR interval spanned zero at each margin. Better acoustic service therefore narrows the latency and channel-demand case for AUV pickup more strongly than it changes the redistribution of gateway transmission energy.

Table 5. Sensitivity to the synthetic acoustic margin. D/H denotes direct priority/hybrid priority; Δ is hybrid minus direct. Values use 10 matched seeds at each margin.

M (dB)	Direct RDR (%)	Hybrid RDR (%)	Δ RDR [95% CI]	HIGH delay D/H (s)	Acoustic D/H (MB)
52	99.81	99.24	−0.57 [−1.35, 0.20]	15.08 / 3.56	4.95 / 4.35
58	99.85	99.24	−0.62 [−1.62, 0.39]	4.82 / 3.30	3.21 / 3.20
64	99.85	99.24	−0.62 [−1.62, 0.39]	4.82 / 3.30	3.21 / 3.20

Fallback timing controls the opposing status cost. Reducing the MEDIUM age gate from 120 to 60 s increased RDR from 98.59% to **98.74%**, reduced mean status misses from 4.67% to **4.17%**, and reduced delivered-status delay from 233.8 to **178.6 s**. The pooled miss ratio fell below the 5% target, although the run-level interval still crossed that threshold. The 60-s setting was the best tested balance; waiting 240 s produced the lowest delivery and the largest status delay and miss ratio.

Table 6. Hybrid sensitivity to the MEDIUM fallback age gate. Intervals are Student 95% intervals over 20 independent runs; pooled ratios provide an event-level audit of the run-level inference.

τ_M (s)	RDR (%)	MEDIUM miss (%) [95% CI]	Pooled miss (%)	MEDIUM delay (s)	MEDIUM expirations
60	98.74	4.17 [1.83, 6.51]	4.62	178.6	51
120	98.59	4.67 [2.18, 7.16]	5.17	233.8	57
240	97.85	7.14 [3.38, 10.89]	7.89	341.3	87

5.3. Continuous Production Exposes a Gateway-Service Bottleneck

Allowing finite traffic to drain measures eventual completion; keeping production active reveals whether the service can repeatedly meet deadlines. The near-parity observed after drain did not persist under continuous demand. Direct priority delivered **99.983%** of records at 10 kB and **99.856%** at 25 kB, and cleared all or nearly all stored data after each patrol cycle. Hybrid priority delivered **91.459%** at both payloads, more than eight percentage points below direct priority. Mature alarm misses remained below 0.54% on average, but mature status misses reached **15.02%** and **15.28%**, well above the 5% reference target.

The terminal state identifies where incomplete work remained. Across the five runs at each payload, hybrid service ended with 1113 records stored at gateways and an empty AUV carry queue; 844 status records had expired. This endpoint pattern localizes the observed shortfall at the gateway-to-pickup stage, where waiting, periodic pickup opportunity, fixed-order sharing, and token-gated fallback shape service. Similar results at 10 and 25 kB reflect the record-count gateway capacity used in this experiment; byte-limited storage remains a hardware-level scaling question.

Table 7. Continuous-arrival service over five complete post-warm-up patrol cycles. Metric entries are run means with 95% Student half-widths over five runs. A/D/S/P are pooled terminal seabed-access losses, deadline expirations, stored records, and pending transactions; every record passed admission.

Service	LOW (kB)	Terminal RDR (%)	Mature HIGH miss (%)	Mature MEDIUM miss (%)	Pooled A/D/S/P
Direct priority	10	99.983 ± 0.012	0.000 ± 0.000	0.015 ± 0.041	1/0/0/3
Direct priority	25	99.856 ± 0.037	0.360 ± 0.268	0.309 ± 0.132	29/0/1/3
Hybrid priority	10	91.459 ± 0.790	0.452 ± 0.685	15.015 ± 3.100	0/844/1113/0
Hybrid priority	25	91.459 ± 0.807	0.539 ± 0.669	15.279 ± 3.317	0/844/1113/0

Cycle-end sampling confirms that the stored data form a persistent operating state rather than a transient snapshot. Direct priority was empty or nearly empty after every cycle. Hybrid priority ended each cycle with approximately 216–224 records at gateways, while both fitted trend intervals spanned zero. The queue does not grow over the five observed cycles because records also expire, yet the stable residue and missed deadlines show that the configured service does not meet the status requirement.

Table 8. Mean stored gateway-plus-AUV records at exact patrol-cycle ends. Trends are run means with 95% Student half-widths over five runs and use the five displayed cycle-end points.

Service	LOW (kB)	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Trend (records/cycle)
Direct priority	10	0.0	0.2	0.0	0.0	0.0	−0.02 ± 0.06
Direct priority	25	0.2	0.6	0.8	0.6	0.2	0.00 ± 0.23
Hybrid priority	10	223.0	215.8	224.0	218.6	222.6	0.20 ± 10.19
Hybrid priority	25	223.0	215.8	224.2	218.8	222.6	0.22 ± 10.14

5.4. Energy Benefits Depend on Patrol Reuse

The energy conclusion changes with the deployment boundary. At the 0.5-mW low-power reference, hybrid service reduced modeled submerged-device communication energy by **80.0%**, from 11.99 to 2.40 kJ. This is a large relative reduction within a narrow local account: 99.66% of the difference arose at the eight gateways. The same active-energy difference produced reductions from 93.1% at 0.1 mW to 4.3% at 50 mW and crossed below 50% near 2.20 mW per nominal position. The percentage therefore depends on the idle-power regime and quantifies gateway communication redistribution; complete platform energy is evaluated separately.

Table 9. Sensitivity of submerged-device energy to idle or hibernation power per nominal submerged position. Values compare hybrid priority with direct priority; 0.5 mW is the low-power reference input.

Idle/hibernation power per position	Direct priority (kJ)	Hybrid (kJ)	Reduction (%)
0.1 mW	10.30	0.71	93.1
0.5 mW	11.99	2.40	80.0
1 mW	14.10	4.51	68.0
2 mW	18.32	8.73	52.3
5 mW	30.98	21.39	31.0
10 mW	52.09	42.50	18.4
20 mW	94.30	84.71	10.2
50 mW	220.93	211.34	4.3

When the patrol exists independently of collection, total incremental communication energy fell by only 7.3%, from 12.89 to 11.95 kJ. The absolute saving was 0.94 kJ, equivalent to an average of 0.040 W over the observation horizon, so a small collection-induced vehicle load removes it. When the full vehicle operation was assigned to collection, hybrid energy was 5.324 MJ compared with 0.0129 MJ for direct priority. Hybrid collection thus has a favorable energy case when it reuses an unchanged inspection patrol; a dedicated collection mission is dominated by vehicle operation.

6. Discussion

Together, the results define a conditional operating boundary. Hybrid pickup accelerates alarms and lowers gateway acoustic demand, whereas continuous production exposes a status-service bottleneck. The balance depends on direct link quality, fallback timing, and whether collection reuses an existing patrol.

6.1. What the Two Observation Horizons Reveal

The finite and continuous experiments answer different deployment questions. When generation stops and queues are allowed to drain, direct and hybrid delivery ratios remain close because both services receive additional time to complete stored work. That experiment measures eventual recovery from a finite episode. Continuous production instead asks whether the architecture can repeat its service cycle while new alarms, status updates, and routine records continue to arrive. The drop from near-complete finite delivery to 91.46% continuous hybrid delivery follows from that change in observation horizon and reveals a capacity and deadline limitation hidden by queue drain.

This distinction defines the main scientific result. Hybrid pickup preserves almost complete delivery for a finite workload while substantially accelerating alarms, whereas sustained demand exposes a status-service capacity limit at the gateways. The two operating modes define an architecture-selection boundary: a campaign that tolerates delayed routine recovery and an installation that requires sustained status deadlines support different deployment choices even when they use the same hardware.

6.2. Mechanisms behind the Trade-off

The three configurations isolate two mechanisms behind the alarm improvement. Priority scheduling reduces urgent delay without changing the direct path; hybrid allocation then removes routine records from that path and produces the additional reduction. This decomposition attributes the first gain to scheduling and the additional gain to AUV-assisted path allocation. The acoustic-margin control confirms the mechanism: as direct links improve, the remaining latency and channel-demand advantage of hybrid pickup shrinks.

The status limitation appears at the gateway-to-pickup stage. Under continuous demand, the AUV carry queue is empty at cycle end while approximately 220 records remain at gateways. Status expiration continues alongside a stable cycle-end residue. This endpoint pattern identifies pickup opportunity, gateway sharing, fallback timing, and fallback reserve as active design variables. The finite fallback sensitivity confirms that this cost is tunable: the 60-s setting gives the best tested status balance, with a pooled miss ratio below the target and a run-level interval overlapping it. This diagnosis complements optimization and learning studies that select routes, links, and schedules within AUV-assisted systems [13–16,31]: the matched baseline identifies where an adaptive policy must create additional value and which outcomes must remain visible.

6.3. Deployment Decision Boundary

These mechanisms and their energy implications define the deployment recommendation. **Direct priority** is the stronger choice when continuous completion, strict status timeliness, or a sufficiently capable acoustic service governs the mission. It also avoids assigning vehicle operation to data collection when no inspection patrol is already planned. **Hybrid priority** becomes attractive when alarms carry much greater operational value than routine-data waiting, when acoustic demand at the gateways must be reduced, and when an AUV already visits the asset for inspection. In that case, the vehicle performs a second useful task without requiring a separate collection mission.

The same boundary resolves the apparently different energy results. The 80% reduction describes modeled communication at submerged equipment under the low-power reference. Once surface and AUV communication are added, reuse of an unchanged patrol yields only a 7.3% incremental saving. Assigning the complete vehicle operation to collection reverses the comparison because propulsion and hotel load dominate. Hybrid collection therefore redistributes work away from inaccessible gateways. Its system-level energy case depends on patrol reuse and the chosen accounting boundary.

6.4. Scope, Validity, and Field Transfer

The evidence applies to a 12.6-km gateway-to-surface scenario with synthetic traffic, a fixed AUV route, 15 surface receivers, record-count gateway storage, and the tested link and fallback settings. Acoustic sensitivity at 52, 58, and 64 dB and idle-power sensitivity from 0.1 to 50 mW show that the main decision changes with link quality and the chosen equipment boundary. The five-run continuous study diagnoses the four tested operating points; a broader capacity map requires additional offered loads, fallback reserves, visit frequencies, gateway-sharing policies, byte-limited buffers, and time-resolved utilization of the carriage and final-transfer stages.

Field transfer requires measured bathymetry, sound-speed profiles, multipath, interference, modem error curves, optical water properties, vehicle power, and the selected surface interface. Recovery, recharge, support-vessel time, and receiver installation also enter a project-level energy and cost account. These measurements calibrate the decision boundary without changing the central test: compare complete direct and hybrid service chains under the same workload and attribute vehicle cost according to the mission that causes it.

7. Conclusions

This matched end-to-end study establishes when AUV-assisted hybrid reporting adds value to a prioritized acoustic service for subsea pipeline monitoring. Direct FIFO and direct priority separate the scheduling effect, while direct priority and hybrid priority separate the effect of changing the routine-data path. Finite traffic followed by drain mea-

sures eventual recovery; continuous production exposes sustained-service capacity. Separate reporting of submerged, incremental-patrol, and dedicated-mission energy connects each energy result to the deployment that produces it.

For the finite reference workload, hybrid pickup reduced alarm delay by **76.1%** and acoustic demand while lowering final delivery by **1.22 percentage points**. That near-parity after drain did not extend to continuous operation: about **15%** of mature status records missed their deadlines, and retained data localized the constraint at the gateways rather than in AUV carriage or final transfer. The modeled submerged-device reduction reached **80.0%** under the low-power reference, but the patrol-reuse saving fell to 7.3%, and vehicle operation dominated under a dedicated mission. Direct priority is the robust choice for continuous completion and status timeliness. Hybrid pickup creates value when rapid alarms, reduced acoustic demand, lower burden at inaccessible gateways, and reuse of an existing inspection patrol govern the deployment decision.

Appendix A Implemented Service, Link, and Energy Functions

This appendix documents the exact simplified functions used by the event-driven evaluation, keeping the main system narrative focused on the deployment study. The equations make distance, payload, pointing, and energy assumptions auditable and define the reference calibration. ALOHA contention, serialization, timeouts, retransmissions, and reassembly remain discrete events. The AquaSim RangePropagation module filters each frame by the transmitting PHY range before MAC collision processing, creating spatial reception neighborhoods. The acoustic probability below then supplies an independent distance-dependent reception draw after successful contention of an addressed frame.

The hybrid status service uses a 1-s scheduler tick, one token of burst capacity, and a nominal network-wide refill of 0.015 record/s. Discrete refill therefore supplies at most one fallback credit every 67 s, or approximately 0.0149 record/s, to eligible MEDIUM records. Reference eligibility begins at age 120 s; the sensitivity study uses 60 and 240 s. One prescribed patrol cycle lasts 17,559.606 s. The continuous ledger is sampled after warm-up at 21,559.606, 39,119.212, 56,678.818, 74,238.424, and 91,798.030 s. Every trend therefore uses five equivalent cycle boundaries.

For an acoustic physical fragment with payload b bytes and geometric distance d in metres, let $d_a = \max(d, 1)$ and let f be in kHz. The implemented absorption, practical-spreading loss, simplified signal-to-noise quantity, and independent-fragment success probability are

$$\begin{aligned}\alpha(f) &= \frac{0.11f^2}{1+f^2} + \frac{44f^2}{4100+f^2} + 2.75 \times 10^{-4}f^2 + 0.003, \\ L_a(d, f) &= 15 \log_{10}(d_a) + \alpha(f) \frac{d_a}{1000}, \\ \gamma_a(d, f) &= 10^{(M-L_a(d, f))/10}, \\ p_a(d, b) &= \left(1 - \frac{1}{2}e^{-\gamma_a(d, f)}\right)^{8b}.\end{aligned}\tag{A1}$$

Here α is in dB/km, L_a and the reference margin M are in dB, and p_a is applied independently to each physical fragment. The first line is the Thorp absorption law, the $15 \log_{10} d_a$ term implements practical spreading, and the final line is the simulator-specific exponential bit-error model inspired by the form used for noncoherent binary FSK [4,33,41]. The quantity γ_a is a simulator-specific signal-to-noise proxy. Payload bytes define the success exponent, while the 41-byte application header contributes to serialization time and en-

ergy. The transmit-power brackets in Table A1 feed the energy ledger independently of γ_a and p_a .

An optical record becomes eligible when the gateway-to-AUV distance is at most d_g . For $d_o = \max(d, 0.1)$, let $U \sim \mathcal{U}(0, 1)$ and $U^* = \max(U, 10^{-12})$. This produces the Rayleigh radial pointing offset r , which is held for the assumed coherence interval. With full divergence angle ϕ , the implemented pointing gain, calibrated line-of-sight signal-to-noise ratio, OOK bit error probability, and whole-record success probability are

$$\begin{aligned} r &= \sigma \sqrt{-2 \ln U^*}, & w(d) &= \max\left(0.01, d_o \tan \frac{\phi}{2}\right), \\ g_p(d, r) &= \exp\left(-\frac{2r^2}{w(d)^2}\right), & \gamma_o(d, r) &= \gamma_0 \left(\frac{d_0}{d_o}\right)^2 \exp(-cd_o + c_0 d_o) g_p(d, r), \\ q_o(d, r) &= Q\left(\sqrt{2\gamma_o(d, r)}\right), & p_o(d, B) &= (1 - q_o(d, r))^{8B}, \end{aligned} \quad (\text{A2})$$

where $Q(x) = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$ and B is the record size in bytes. The $1/d^2$ and Beer-Lambert terms preserve geometric spreading and extinction, while the Gaussian transverse error gives a Rayleigh radial offset and the last line gives the direct-detection OOK mapping [5,6,42,43]. The normalization (d_0, γ_0) fixes the aligned clear-water reference near a 10^{-5} OOK bit error rate [44,45]. d_0 sets the link-budget scale, while $d_g = 30$ m defines transfer eligibility. The success probability is fixed at transfer start, and delivery additionally requires the AUV to remain within d_g when serialization completes.

The hardware energy ledger uses serialized time at active power. If n_w is the acoustic wire size in bytes, including the 41-byte header, and R_a and R_o are the acoustic and optical rates, the charged event energies and the vehicle power model are

$$\begin{aligned} E_{a,\text{tx}}(d, n_w) &= P_{a,\text{tx}}(d) \frac{8n_w}{R_a}, & E_{a,\text{rx}}(n_w) &= P_{a,\text{rx}} \frac{8n_w}{R_a}, \\ E_{o,\text{success}}(B) &= (P_{o,\text{tx}} + P_{o,\text{rx}}) \frac{8B}{R_o}, & E_{\text{AUV}}^{\text{op}} &= (5.23 + 45.78v^3)T. \end{aligned} \quad (\text{A3})$$

The constant-plus-cubic vehicle term represents hotel load plus the cubic speed dependence expected from steady hydrodynamic drag [36,37]; its two coefficients define the reference vehicle scenario. Acoustic transmit energy accrues for every attempt, and receive energy accrues after successful reception. Optical energy is split between the gateway and AUV and accrues on serialized successful contacts. An idealized 16-byte acoustic acknowledgement uses the same transmit and receive functions and represents an immediate successful reverse path. These choices define the event-energy boundary.

Table A1. Active parameters for the implemented service, link, and energy functions.

Component	Active value	Interpretation and anchor
Acoustic success draw	$f = 10\text{ kHz}$; $M = 52\text{ dB}$ reference, with 58 and 64 dB sensitivity; spreading coefficient 1.5	Synthetic margin with Thorp absorption, independent of site-noise calibration [4,33].
Acoustic framing	$R_a = 31.2\text{ kbit/s}$; 4096-byte maximum fragment payload; 41-byte header	Rate anchor from the published S2C modem datasheet; the header is included in airtime and energy [34].
Acoustic service	One initial attempt plus three retries; 12-s timeout; backoff uniform from 0.5 to 1.5 s; 16-byte acknowledgment	Range-filtered reception and spatial collision neighborhoods; the local budget resets after requeue, with record deadline as the terminal retry limit for both gateway and AUV surface service.
Acoustic energy	$P_{a,\text{tx}} = 5.5, 8, 18, \text{ or } 40\text{ W}$ for $d \leq 250$, $d \leq 500$, $d \leq 1000$, or $d > 1000\text{ m}$; $P_{a,\text{rx}} = 0.8\text{ W}$	Transmit brackets use the S2C power specification; 0.8-W receive power is a scenario input. The success draw uses the independent margin M [34].
Optical channel	$d_g = 30\text{ m}$; $c = c_0 = 0.151\text{ m}^{-1}$; $d_0 = 68.5\text{ m}$; $\gamma_0 = 9.10$	Clear-ocean Beer–Lambert normalization set to an OOK bit error rate near 10^{-5} at (d_0, c_0) when $g_p = 1$, corresponding to zero pointing offset [5,44,45].
Optical pointing	$\phi = 30\text{ degrees}$; $\sigma = 0.75\text{ m}$; 60-s coherence interval	Scenario inputs for the simplified Gaussian/Rayleigh alignment process [42,43].
Optical energy	$R_o = 2.5\text{ Mbit/s}$; $P_{o,\text{tx}} = 15\text{ W}$; $P_{o,\text{rx}} = 10\text{ W}$	Rate and active-power anchors from BlueComm 200 [35].
Vehicle operation	$v = 6.087\text{ km/h}$; $T = 23,450\text{ s}$; $P_{\text{AUV}}(v) = 5.23 + 45.78v^3\text{ W}$ for v in m/s	Scenario fit consistent with cubic steady-drag scaling; platform-specific calibration uses measured vehicle power [36,37].
Idle reference	90 mW aggregate across submerged communication endpoints	Hibernation-scale 0.5-mW allowance for each of 180 sensor positions, redistributed across 188 submerged endpoints; evaluated from 0.1 to 50 mW [46].

Author Contributions: Conceptualization, A.M.D., N.R. and J.A.; methodology, A.M.D.; software, A.M.D.; validation, A.M.D.; formal analysis, A.M.D.; investigation, A.M.D.; writing, original draft preparation, A.M.D.; writing, review and editing, N.R. and J.A.; supervision, N.R. and J.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: A supplementary reproducibility package containing the simulator source, pinned build environment, experiment workflows, analysis code, execution logs, per-run time series, and data underlying the reported results accompanies the manuscript. The development repository is available at <https://github.com/Aissata98/ns3-new>.

Conflicts of Interest: The authors declare no conflicts of interest.

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